

LAB # 04b

Implementation of math class with various methods in java

Objective

To get familiarity with the math class with various methods in java.

Lab Goals

To provide working insight into the usage of math class in java programming:

- a. Demonstrate the implementation of math class.
- b. Understand the purpose and usage of the Math class in Java.
- c. Learn about the various mathematical functions and constants provided by the Math class.

Math Function

Math Class in Java contains certain methods to perform different numeric operations, for instance, *exponential, square root, logarithmic, trigonometric* functions.

Method	Description
Math.abs()	It will return the Absolute value of the given value.
Math.max()	It returns the Largest of two values.
Math.min()	It is used to return the Smallest of two values.
Math.round()	It is used to round of the decimal numbers to the nearest value.
Math.sqrt()	It is used to return the square root of a number.
Math.cbrt()	It is used to return the cube root of a number.
Math.pow()	It returns the value of first argument raised to the power to second argument.

The Java Math class has many methods that allows you to perform mathematical tasks on numbers.

Math.max(x,y)

The Math.max(x,y) method can be used to find the highest value of x and y.

Example

```
public class Main {  
    public static void main(String[] args) {  
        System.out.println("The Max Value is :: " + Math.max(5, 10));  
    }  
}
```

Output

```
The Max Value is :: 10
```

Math.min(x,y)

The Math.min(x,y) method can be used to find the lowest value of x and y.

Example

```
public class Main {  
    public static void main(String[] args) {  
        System.out.println("The Min Value is :: " + Math.min(5, 10));  
    }  
}
```

Output

```
The Min Value is :: 5
```

Math.sqrt(x)

The Math.sqrt(x) method returns the square root of x.

Example

```
public class Main {  
    public static void main(String[] args) {  
        System.out.println("The Square root of is :: " + Math.sqrt(64));  
    }  
}
```

Output

```
The Square root of is :: 8.0
```

Math.abs(x)

The Math.abs(x) method returns the absolute (positive) value of x.

Example

```
public class Main {  
    public static void main(String[] args) {  
        System.out.println("The absolute value of -4.7 is :: " + Math.abs(-4.7));  
    }  
}
```

Output

```
The absolute value of -4.7 is :: 4.7
```

Random Numbers

Math.random() returns a random number between 0.0 (inclusive), and 1.0 (exclusive) but never exactly 1.0).

Example

```
public class Main {  
    public static void main(String[] args) {  
        System.out.println("The random number is :: " + Math.random());  
    }  
}
```

Output

```
The random number is :: 0.3497740931649802
```

Method 1 (custom start and end range)

To generate a random number between a specific starting range and end range using Math.random():

General Formula

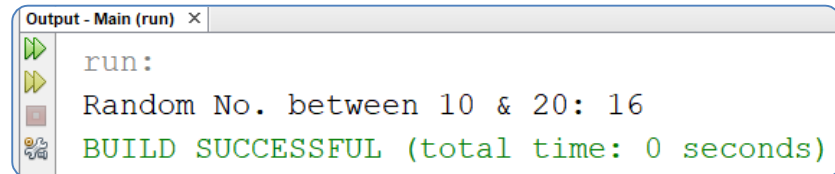
```
int randomNumber = (int)(Math.random() * (endRange - startRange + 1)) + startRange;
```

Example: Generate a Random Number Between 10 and 20

```
package main;  
public class Main {  
    public static void main(String[] args) {  
        int start = 10; // Custom start range  
        int end = 20;   // Custom end range
```

```
int randomNum = (int)(Math.random() * (end - start + 1)) + start;
System.out.println("Random No. between " + start + " & " + end + ": " + randomNum);
}
}
```

Output



```
Output - Main (run) x
run:
Random No. between 10 & 20: 16
BUILD SUCCESSFUL (total time: 0 seconds)
```

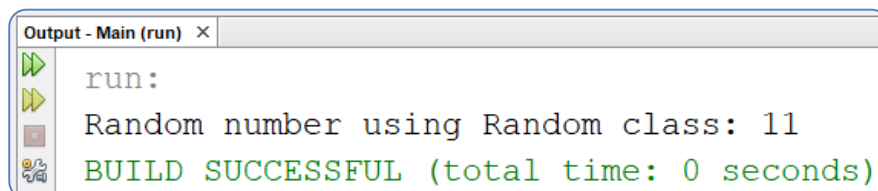
Code working Explanation:

- Math.random() takes the range $\rightarrow 0.0$ to $0.999\dots$
- When we apply custom range **end - start + 1** $\rightarrow 20 - 10 + 1 = 11$.
Note: The +1 ensures the **end** value (**20**) is included.
- Math.random() * 11 $\rightarrow 0.0$ to $10.999\dots$
- (int) **truncation** (explicit type casting) $\rightarrow 0$ to 10 (drops decimal).
- + **start (10)** $\rightarrow 10$ to 20

Method 2 (Using Random class)

```
package main;
import java.util.Random;
public class Main {
    public static void main(String[] args) {
        Random random = new Random();
        // random number between 1 and 25
        int randomNumber1 = random.nextInt(25) + 1;
        System.out.println("Random number using Random class: " + randomNumber1);
    }
}
```

Output



```
Output - Main (run) x
run:
Random number using Random class: 11
BUILD SUCCESSFUL (total time: 0 seconds)
```

Exercise

- A. Write a program that generates numbers randomly between 97 to 122 and convert that number to their corresponding character (**Condition:** don't use looping).
- B. Write a program that calculates the hypotenuse of a right-angled triangle using `Math.sqrt()` and the Pythagorean theorem ($a^2 + b^2 = c^2$). Take **a** and **b** as user input.
- C. Write a program to find the **Compound Interest** using **Math.pow()**:

Formula:

$$A = P \times \left(1 + \frac{r}{n}\right)^{n \times t}$$

where:

A = final amount

P = principal amount

r = annual interest rate

n = number of times interest is compounded per year

t = time in years

- D. Write a program that calculates the Euclidean distance between two points (x1,y1) and (x2,y2) using **Math.sqrt()** and **Math.pow()**.

Formula:

$$\text{Distance} = \sqrt{(x2 - x1)^2 + (y2 - y1)^2}$$